Great—this is where things get serious. I'll tailor this to **your profile**:

- 18 years engineering + architecture
- Strong systems thinking
- Already exploring LLM internals 👉 So we **skip beginner fluff** and go straight to **research-grade execution**

🚀 12-Month AI Research Roadmap (Week-by-Week)

🎯 End Goal (Non-negotiable)

By Month 12 you should have:

- 2 research-grade projects
- 1 arXiv-style paper
- 1 strong specialization (LLM + Systems)
- Deep understanding of transformer internals + infra

📅 MONTH 1–2 (Weeks 1–8) → Mathematical + Transformer Foundations

Week 1

- Linear algebra refresh:
  - vectors, matrices, dot product
- Implement from scratch:
  - matrix multiplication (NumPy)

Week 2

- Eigenvalues, SVD
- Why embeddings work mathematically

Week 3

- Probability basics:
  - distributions, expectation, variance
- Implement:
  - sampling functions

Week 4

- Optimization:
  - gradient descent, loss surfaces
- Implement:
  - gradient descent manually

Week 5

- Neural networks from scratch:
  - forward pass, backprop
- Build:
  - 2-layer NN (NumPy only)

Week 6

- Activation functions, initialization
- Debug vanishing gradients

Week 7

- Softmax + cross entropy (deep understanding)
- Implement classification model

Week 8

- Attention mechanism (core week ⚠️)
- Implement:
  - scaled dot-product attention from scratch

📅 MONTH 3–4 (Weeks 9–16) → Build GPT from Scratch

Week 9

- Transformer architecture (encoder vs decoder)
- Focus: decoder-only

Week 10

- Implement:
  - token embeddings
  - positional encoding

Week 11

- Multi-head attention (full implementation)

Week 12

- Feed-forward network + residual connections

Week 13

- Stack full transformer block
- Build mini-GPT architecture

Week 14

- Training loop:
  - batching, loss, optimizer

Week 15

- Train on small dataset
- Debug instability

Week 16

- Add:

- temperature sampling
- text generation

👉 OUTPUT:

- Your own GPT repo (no frameworks except tensor ops ideally)

📅 MONTH 5–6 (Weeks 17–24) → Systems + Scaling (Your Strength Zone)

Week 17

- GPU basics:
  - CUDA conceptually
  - memory vs compute

Week 18

- PyTorch internals:
  - autograd
  - computational graph

Week 19

- Profiling:
  - bottlenecks in training

Week 20

- Mixed precision training (FP16)

Week 21

- Distributed training:
  - data parallelism

Week 22

- Model parallelism
- Understand large model training constraints

Week 23

- Inference optimization:
  - KV cache
  - batching

Week 24

- Build:
  - simple inference server (LLM API)

👉 OUTPUT:

- "Mini OpenAI-like inference system"

📅 MONTH 7–8 (Weeks 25–32) → Paper Reading + Reproduction

Week 25

- Read first serious paper:
  - OpenAI GPT paper
- Break into components

Week 26

- Reproduce part of paper:
  - tokenizer / architecture

Week 27

- Read:
  - attention improvements paper

Week 28

- Implement variation

Week 29

- Study:
  - scaling laws

Week 30

- Experiment:
  - train different sizes → observe loss

Week 31

- Read:
  - RLHF basics

Week 32

- Implement:
  - simple reward model

👉 OUTPUT:

- 2–3 “paper reproduction” repos

📅 MONTH 9–10 (Weeks 33–40) → Research Project (Core Phase)

Week 33–34

Choose YOUR research problem: 👉 Suggested for YOU:

- Efficient transformer (speed/memory)
- Better inference architecture

- LLM system design

Week 35–36

- Literature review (deep dive)
- Identify gap

Week 37–38

- Prototype solution

Week 39–40

- Run experiments
- Compare vs baseline

👉 OUTPUT:

- Research-grade project #1

📅 MONTH 11 (Weeks 41–44) → Paper Writing

Week 41

- Structure:
  - Abstract
  - Problem
  - Method

Week 42

- Add experiments + results

Week 43

- Add diagrams + clarity

Week 44

- Final draft

👉 Publish on:

- arXiv

📅 MONTH 12 (Weeks 45–52) → Positioning + Advanced Work

Week 45–46

- Start project #2 (faster iteration)

Week 47–48

- Improve first paper (based on feedback)

Week 49–50

- Build strong GitHub portfolio

Week 51–52

- Apply / network:
  - Research Engineer roles
  - Collaborate with researchers

🧠 Weekly Time Allocation (Realistic)

Since you're experienced:

- 2 hrs/day weekdays
- 5–6 hrs weekends

🔥 Critical Rules (Don’t Ignore)

1. Always Build

If you only read: ❌ You fail

If you implement: ✅ You become researcher

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2. Always Write

- Maintain research notes
- Write blogs

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3. Always Question Papers

Ask:

- Why this works?
- Can I improve this?

⚡ What Makes This "PhD-Level"

You are replicating:

- Coursework → Months 1–4
- Lab work → Months 5–10
- Thesis → Months 9–11
- Publication → Month 11–12

💡 Tailored Advice for YOU

Given your background, your unfair advantage is:

👉 **AI Systems + Architecture**

So don't compete on:

- basic ML models ❌

Compete on:

- LLM infra
- scalability

- scalability
- performance
- real-world deployment

👉 This is where even PhDs struggle.

🚀 If you want next step

I can:

- Give **3 research problems** that can realistically get you published
- Review your **weekly progress like a PhD advisor**
- Help you **turn your first project into a paper**

Just tell me: 👉 *"Give me research problems"*